

Teorema de Fubini

Teorema 1 (de Fubini). Sean (X_1, Σ_1, μ) y (X_2, Σ_2, λ) espacios de medida σ -finitos.

Sea $f: X_1 \times X_2 \longrightarrow \mathbb{R}$ una función $\Sigma_1 \otimes \Sigma_2$ -medible. Si definimos

$$\begin{aligned} \varphi: X_1 &\longrightarrow \mathbb{R} & \psi: X_2 &\longrightarrow \mathbb{R} \\ x &\longmapsto \int_{X_2} f(x, y) \, d\lambda(y) & y &\longmapsto \int_{X_1} f(x, y) \, d\mu(x), \end{aligned}$$

$$\implies \begin{cases} (1) \ f \geq 0 \implies \varphi \text{ es } \Sigma_1\text{-medible y } \psi \text{ es } \Sigma_2\text{-medible.} \\ (2) \ f \in \mathcal{L}^1(\mu \otimes \lambda) \implies \varphi \in \mathcal{L}^1(\mu) \text{ y } \psi \in \mathcal{L}^1(\lambda). \end{cases}$$

Además, en cualquiera de estos dos casos, se cumple que

$$\int_{X_1 \times X_2} f(x, y) \, d(\mu \otimes \lambda) = \int_{X_1} \varphi(x) \, d\mu(x) = \int_{X_2} \psi(y) \, d\lambda(y).$$

Referenciado en

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